Actor-Critic Methods

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$$\begin{cases} \text{Actor: policy update} \\ \text{Critic: policy eval / value estimation.} \end{cases}$$

Simplest AC

Algo (QAC)

step t:

$$\pi(a|s_t,\theta_t) \Rightarrow a_t \xrightarrow{\text{observe}} r_{t+1}, s_{t+1}$$

$$\pi(a|s_{t+1},\theta_t) \Rightarrow a_{t+1}$$

Critic (TD, Sarsa):

$$w_{t+1} = w_t + \alpha_w \left[r_{t+1} + \gamma q(s_{t+1}, a_{t+1}, w_t) - q(s_t, a_t, w_t) \right] \nabla_w q(s_t, a_t, w_t)$$

Actor (PG):

$$\theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \ln \pi(a_t|s_t,\theta_t)\, q(s_t, a_t, w_{t+1})$$

Advantage AC

since

$$E_{S\sim\eta, A\sim\pi}\left[\nabla_\theta \ln \pi(A|S,\theta_t)\, b(S)\right] = \sum_{s\in S} \eta(s)\, b(s)\, \nabla_\theta \sum_{a\in A} \pi(a|s,\theta_t) = 0$$

therefore

$$\nabla_\theta J = E_{S\sim\eta, A\sim\pi}\left[\nabla_\theta \ln \pi(A|S,\theta_t)\left(q_\pi(S,A) - b(S)\right)\right]$$

but

$$b(s) \longrightarrow Var\left[\nabla_\theta \ln \pi(A|S,\theta_t)\, b(S)\right]$$

want min Var:

$$b^*(s) = \frac{E_{A\sim\pi}\left[\cancel{\|\nabla_\theta \ln \pi(A|S,\theta_t)\|^2}\, q(S,A)\right]}{E_{A\sim\pi}\left[\cancel{\|\nabla_\theta \ln \pi(A|S,\theta_t)\|^2}\right]} \longrightarrow V_\pi(s)$$

let $b(s) = V_\pi(s)$,

then

$$\theta_{t+1} = \theta_t + \alpha E\left[\nabla_\theta \ln \pi(A|S,\theta_t)\, \delta_\pi(S,A)\right]$$

where

$$\delta_\pi(S,A) \triangleq q_\pi(S,A) - V_\pi(S) \quad \text{advantage function}$$

$$\Rightarrow \theta_{t+1} = \theta_t + \alpha \left(\frac{\overset{\text{exploit}}{\delta_t(s_t,a_t)}}{\underset{\text{explore}}{\pi(a_t|s_t,\theta_t)}} \right) \nabla_\theta \pi(a_t|s_t,\theta_t)$$

Algo (A2C)

step t:

$$\pi(a|s_t,\theta_t) \Rightarrow a_t \xrightarrow{\text{observe}} r_{t+1}, s_{t+1}$$

TD error:

$$\delta_t = r_{t+1} + \gamma v(s_{t+1}, w_t) - v(s_t, w_t)$$

Critic:

$$w_{t+1} = w_t + \alpha_w \delta_t \nabla_w v(s_t, w_t)$$

Actor:

$$\theta_{t+1} = \theta_t + \alpha_\theta \delta_t \nabla_\theta \ln\pi(a_t|s_t,\theta_t)$$

Off-policy AC

importance sampling:

$p_1(\theta)$ ✓ $p_0(\theta)$ ✗ Q: $E_{x\sim p_0}[X] = ?$

Solve: $E_{x\sim p_0}[X] = \sum_x p_0(x)x = \sum_x p_1(x)\underbrace{\frac{p_0(x)}{p_1(x)}x}_{=f(x)} = E_{x\sim p_1}[f(x)]$

then, let

$$\bar{f} = \frac{1}{n}\sum_{i=1}^{n} f(x_i), \quad x_i \sim p_1$$

$$\Rightarrow E_{x\sim p_1}[\bar{f}] = E_{x\sim p_1}[f(x)] = \frac{1}{n}\sum_{i=1}^{n} f(x_i)$$

$\Rightarrow$ × importance weight: $\frac{\pi(a_t|s_t,\theta_t)}{\beta(a_t|s_t)}$

stochastic

Deterministic AC (DPG)

$$a = \mu(s,\theta) \triangleq \mu(s)$$

$$\nabla_\theta J(\theta) = E_{s\sim\rho_\mu}[\nabla_\theta \mu(s)(\nabla_a q_\mu(s,a))|_{a=\mu(s)}]$$

$$\Rightarrow \text{GA}: \theta_{t+1} = \theta_t + \alpha_\theta \nabla_\theta \mu(s_t,\theta_t)(\nabla_a q(s_t,a,w_{t+1}))|_{a=\mu(s_t)}$$

Off-policy.

on-policy: β: μ+noise